

02/07/2025



General Motors

McCombs School of Business

University of Texas at Austin

Week 03
Capstone Team 8

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Executive Summary

1. Updated Project Plan showcasing 13 weeks of progression with broad sections, milestones and team assignments
2. Review of target data and KPIs to finalize business outcomes
3. Progress on finalizing data sources and join criteria for modelling dataset
4. Setup of Github Repository for Documentation
5. Follow up questions and requests for latest numbers
6. Feedback

1. Project Plan

Powering the future through smart EV expansion

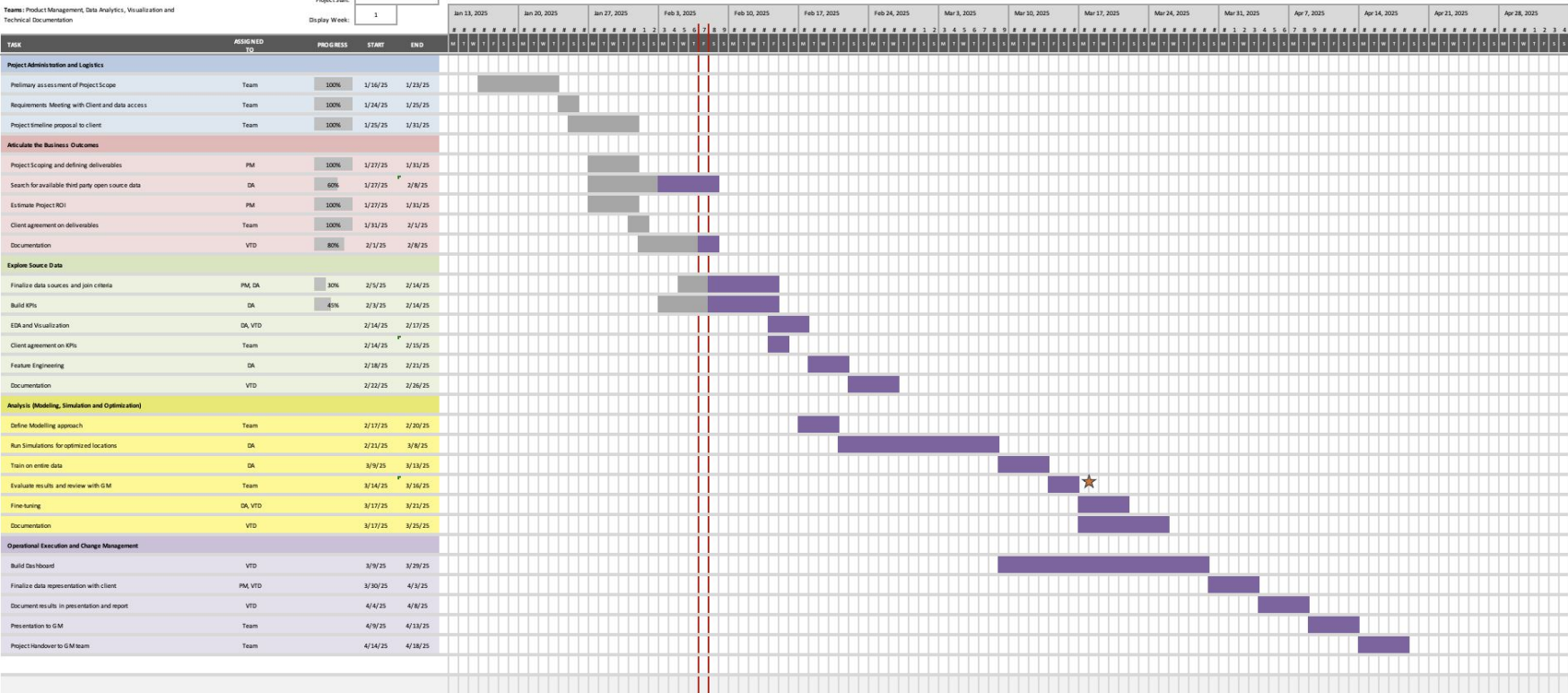
Capstone Team 8: Muhammad Ibrahim, Sarah Lee, John Hwang, Rohan Giri, Varsha Manju Jayakumar, Ammar Mustafa

General Motors

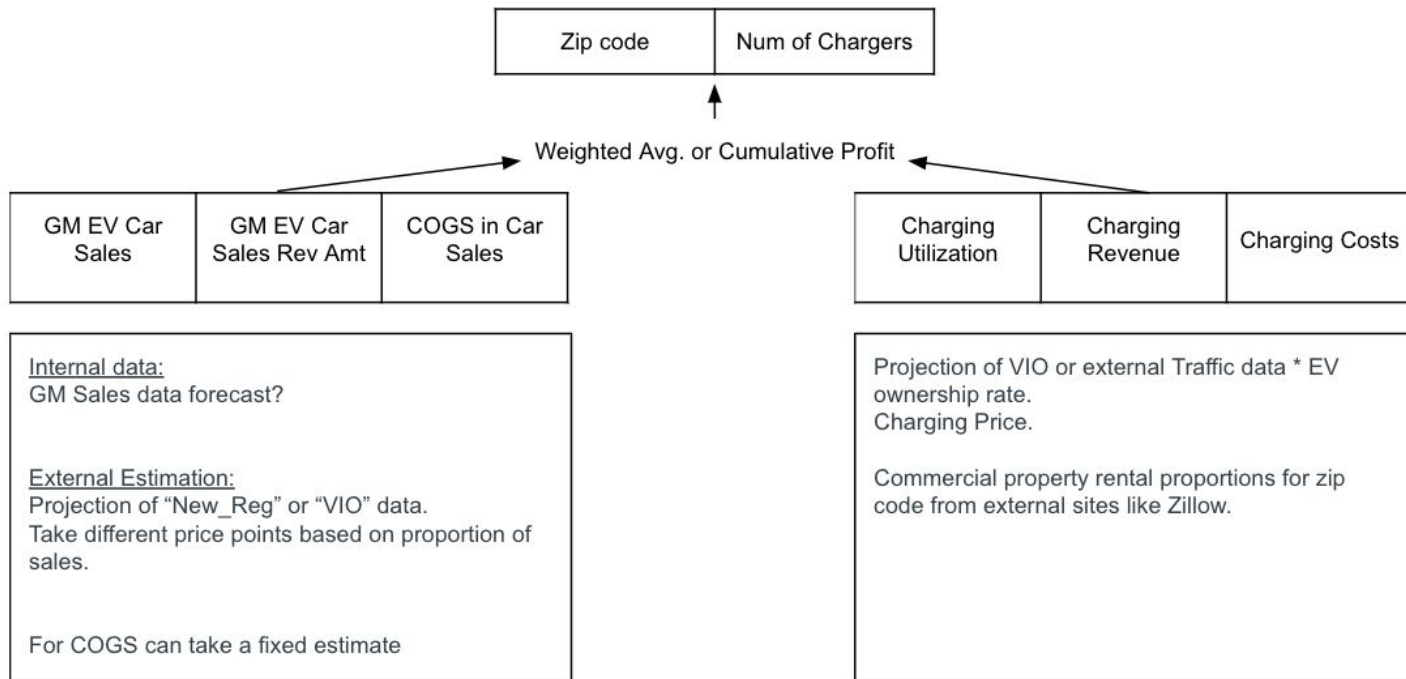
Teams: Product Management, Data Analytics, Visualization and Technical Documentation

Project Start: Thu, 1/30/2025

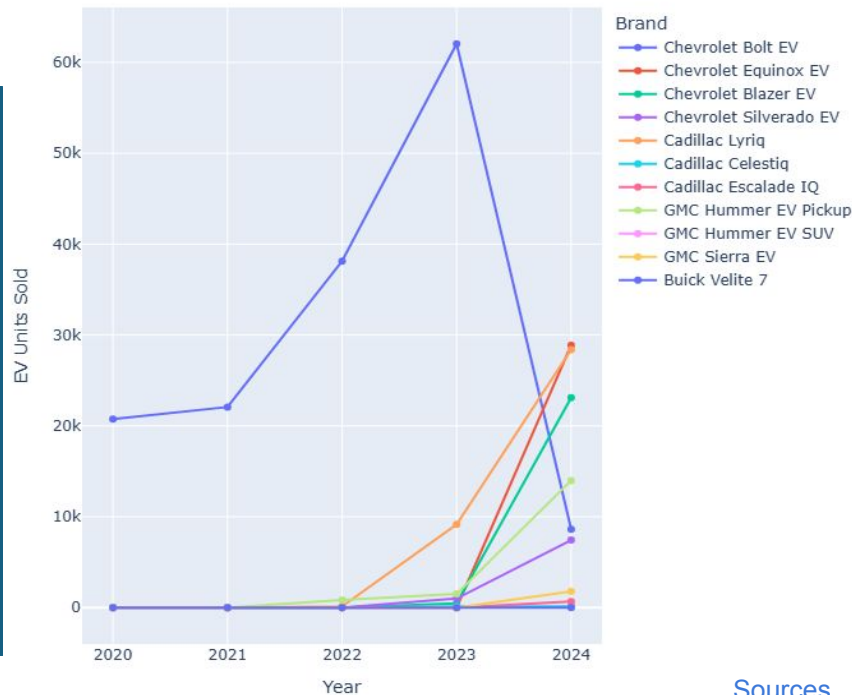
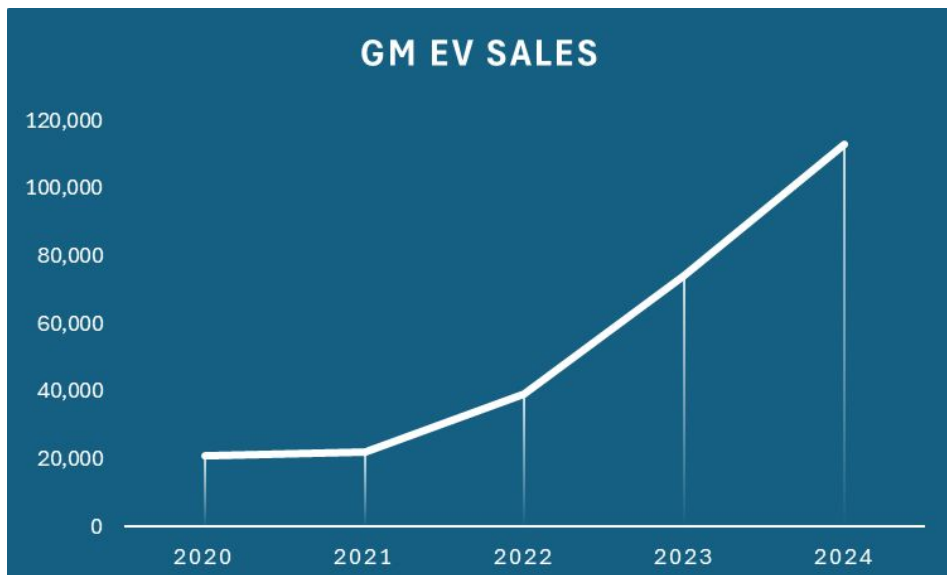
Display Week: 1



Building KPIs



GM EV Car sales



GM EV Car Revenue

Car Sales * (Base Price + Upgrades + Dealership margin)

Model	2020	2021	2022	2023	2024
Bolt EV	\$43,735	\$41,700	\$36,165	\$30,695	\$32,000
Bolt EUV	\$43,995	\$43,995	\$43,995	\$34,000	\$34,000
Equinox EV	0	0	0	0	\$48,000
Blazer EV	0	0	0	0	\$66,000
Silverado EV	0	0	0	0	\$80,000
Lyriq	0	0	\$73,315	\$73,315	\$68,890
Celestiq	0	0	0	0	\$400,000
Escalade IQ	0	0	0	0	\$150,000
Hummer EV Pickup	\$110,295	\$118,039	\$110,295	\$129,415	\$132,895
Hummer EV SUV	\$129,415	\$129,415	\$129,415	\$115,000	\$129,415
Sierra EV	0	0	0	0	\$80,000

Market Growth & Sales Performance

- Total Sales Growth:
 - 2020: 20,754 units
 - 2024: 113,047 units
 - 445% growth over 5 years**

Top Performing Models (2024)

- Cadillac Lyriq: 28,402 units at \$68,890 = **\$1.96B**
- Chevrolet Equinox EV: 28,874 units at \$48,000 = **\$1.39B**
- GMC Hummer EV Pickup: 13,993 u at \$132,895 = **\$1.86B**

Total Combined Revenue from EV Sales : **\$14.67 B**

[Calculation Link](#)

Car sales: Cost

- Capital IQ did not contain explicit financial information for EV specific categories
- Key cost components for an average \$40,000 EV (from Tesla and BYD averages):
 - Battery - \$8,500 (for average vehicle with 65 kWh battery * \$130/kWh in 2024)
 - Chassis, body, powertrain - \$10,000 - \$12,000
 - Labor - \$1,500 - \$2,500
 - Misc - \$2,000 - \$4,000

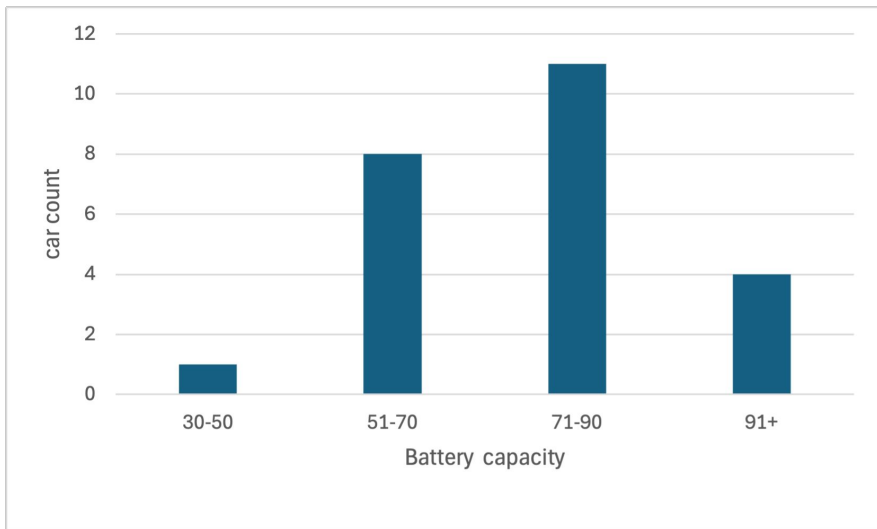
Total COGS (fixed) ~ \$25,500 or ~ 65% of selling price of car

Charger: Revenue

Total revenue = Annual Average Daily Traffic (AADT) * EV rate * Price per unit
* Average car capacity

- AADT → Obtained on zip code level
- EV rate: 6.1 million vehicles registered → 39,353 EVs → 0.65% (TN.gov)
- Avg cost of DC fast chargers in TN: \$0.42 per kWh (American Automobile Association)
- Average capacity: 65.35 kWh

EV Battery Capacity Distribution Analysis



Battery Capacity Categories

- **30-50 kWh:** Smaller capacities (e.g., hybrids, compact EVs).
 - **51-70 kWh:** Moderate capacities (mid-range EVs).
 - **71-90 kWh:** Larger capacities (long-range EVs)
 - **91+ kWh:** Premium or advanced EVs (luxury vehicles).
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- **Average battery capacity** = Total energy/ No. of vehicles = **65.35 kWh**

Charger: Cost

- **Station Rent Costs:**

- Urban: \$13.00, Suburban: \$7.70, Rural: \$3.60
- **Excel**
 - Classified zip codes into rural, suburban and urban
 - Found annual rent of gas stations based on the classified zip code
 - Assumed 10,000 sqft for gas station size
 - Got \$/sqft of gas stations based in zip

- **Ongoing Infrastructure & Maintenance Cost (Fixed cost)**

- Maintenance & Cleaning - \$400 - \$1,000/ year
- Inspections & Repairs: \$500 - \$2,000/ year
- Replacements: \$2,000 - \$5,000/ 5 years (~\$400 - \$1,000/ year)
- HVAC & Cooling Specific to DC: \$500 - \$1,500/ year

Average Infrastructure & Maintenance Cost: \$3,400/ year

Progress on Data Merging

- GM data → Zip code level
- Traffic data → (details on the next slide)
- Charging stations → granularity on latitude/longitude level
- EV_reg → transformed into count of each vehicle make per county

Assigning Accurate ZIP Codes to Tennessee Traffic Routes

- **Started with the Traffic Dataset:** The traffic dataset lacked ZIP codes, so it was supplemented with a statewide dataset containing routes and multiple latitude/longitude points along those routes.
- **Calculated Representative Coordinates:** Averaged the latitude and longitude for each route to determine a single representative location for that route.
- **Matched ZIP Codes to Coordinates:** Using a different dataset of ZIP codes with latitude and longitude, it was filtered to include only Tennessee ZIP codes (37010–38589) and matched each route's representative location to the closest valid ZIP code.
- **Validated ZIP Codes by County:** Cross-referenced the assigned ZIP codes with an official county-to-ZIP mapping. Rows with ZIP codes that didn't belong to their respective counties were removed.
- **Resulting in our routes having zip codes assigned.**

Feedback